

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3		(i)	Similarity: They both have 2 [or the same number of] protons / 4 [or same number of] nucleons (1) Not: Both have a nucleus / have the same nucleus – treat as neutral Difference: <u>Helium atom</u> has [2] electrons (orbiting the nucleus)/ <u>alpha</u> has no electrons (1) Not: Reference to charge or size – treat as neutral	2			2		
		(ii)	206 (1) 82 (1)		2		2		
		(iii)	Po has an imbalance between the numbers of protons and neutrons [accept: has more neutrons than protons] / too many neutrons / it has an unstable <u>nucleus</u>	1			1		
		(iv)	<u>Alpha</u> particles are [accept: the radiation is] very ionising (1) Which kills cells / mutates cells or DNA [accept: causes cancer] (1) Not: radioactive / poisonous		2		2		
			Question 3 total	3	4	0	7	0	0